

SinkInSync: A Brainwave Synchronization Experience in VR Towards Enhancing Social Connectedness

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ABSTRACT

SinkInSync presents the design and prototype of a VR-based cross-person EEG neurofeedback platform. This generative VR platform uses one user's brainwave data to procedurally render 3D scenes and passively displays visual cues that are synchronized with the real-time brainwave frequency to another user. With the platform, we aim to explore the potential of VR as an avenue for augmenting cognitive and emotional social connectedness in remote interactions via externally-induced brainwave synchronization between pairs of individuals.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Virtual reality**;

KEYWORDS

Biofeedback VR, Brain-computer interaction, Social connectedness, Emotional wellbeing, VR immersive gaming

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1 INTRODUCTION AND RELATED WORK

People tend to synchronize heart rates, breathing and neural rhythms during intimate physical contact and emotional exchange. This synchronization in turn enhances our mutual feelings of social connectedness [McCraty, 2017]. In a world of rising remote interactions, we have less chance for in-person interactions that precondition socially-induced synchronization. We were motivated by previous research that uses VR as an alternative method to augment cognition, such as [Amores et al., 2019] that uses EDA data to modify the VR environment in real-time to help with relaxation. Other examples include using VR for social phobia [Gebara et al., 2015] and empathy [Kiltner et al., 2012]. We were also inspired by

previous work that aims to use external cues to influence behavior, such as using light changes to influence breathing and increasing focus [Ghandeharioun and Picard, 2017], and were informed by previous research that has discovered the capability of brainwaves to synchronize with external stimuli [Will and Berg, 2007]. In our work, we used visual rhythms to influence the users' brainwave in a cross-person VR environment.

2 DESIGN AND IMPLEMENTATION

Imagine a child with autism who experiences difficulty in socializing verbally and physically. SinkInSync pairs this child with an outgoing child and offers a novel interactive modality that does not require speaking by immersing them in VR environments generated by each other's brainwaves in real time. We designed this cross-person neurofeedback experience with rules that associate brainwave data with 3D scene elements in a fictional VR landscape.

Based on common narratives about the psychology of colors, we assigned a main hue to each dominant brainwave band to strengthen their influence towards the corresponding mental states. Higher activeness would increase warm colors while higher relaxation would increase cold colors in the scene elements. The lanterns overhead synchronize their pulsating light with the dominant brainwave frequency. The large, opaque lanterns embody the color of the dominant wave band while the smaller, translucent ones take on that of the secondary and tertiary ones. The dust-like particles floating in the air come various colors, creating an ambience of dispersed and nuanced thoughts that exist beyond our dominant mental state. As the user wanders in this landscape controlled by the other user's brainwave, they can subtly observe their own dominant brainwave in the color of the fog flowing in the far backdrop and experience a signal if the two synchronize.

Each user's brainwave data is captured by a Muse EEG headband. We categorized and transmitted the band types along with the dominant frequency into Unity 3D via the OSC server and built-in WebSocket library. In Unity, we processed and mapped the data to CG properties. The dominant band type was used to control the color properties of the scene elements, and the dominant frequency was used to update the time multiplier of the glowing shader that renders the mesh objects, the frequency of a Perlin noise operator in the VFX graph, and the lifetime and duration properties of a particle system.

3 DISCUSSION

In this work SinkInSync, the use of VR is successful in creating a strong embodied experience that strengthens the nudge for synchronization and connection in remote interactions. Using commercially

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available tools such as Muse headband and Oculus VR headsets, the platform promises to be affordable and accessible to be used rapidly by the public. The current prototype and user testing proves the potential of SinkInSync to be useful in different real-life situations with an impact on social connectedness and emotional wellbeing, such as relationship therapy, autism therapy, medication, past-self immersion, and novel interactive games. At a high level the project promotes reflections around the potential of emerging technologies like VR in fostering tele-present interpersonal connections without the common interactive modalities like speech and how the usage of biometric data could add to the channels of communication and help improve accessibility.

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